

# HARSH VICTOR CHALLA

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## EDUCATION

### Arizona State University, Tempe, AZ, USA

Master of Science in Computer Science

August 2023 - May 2025

GPA: 4/4

### Visvesvaraya Technological University, Bengaluru, KA, India

Bachelor of Engineering in Computer Science

August 2019 - May 2023

GPA: 3.6/4

**Relevant Courses:** Data Structures, Big Data Analytics, Artificial Intelligence and Machine Learning, Data Mining, Data Communication, Distributed Database Systems, Data Visualization, Statistical Machine Learning, Cloud Computing.

## EXPERIENCE

### Arizona State University

Business Data Analyst

November 2023 - Present

Tempe, AZ, USA

- Designed and implemented Python scripts to automate the formatting of accounts data, reducing processing time by 97.22%, from 12 hours to just 20 minutes, while improving data accuracy and consistency across reports.
- Collaborated with end users to develop Amazon Redshift data models by integrating spending records from PeopleSoft and Workday, providing capital expenditure insights for the company.

### Tequed Labs

Machine Learning Intern

September 2022 - September 2022

Bengaluru, KA, India

- Led data cleaning and preprocessing initiatives for the house price prediction project, utilizing Python to handle missing values, outlier detection, and feature engineering, resulting in a model with 20% improved pricing accuracy in a tight timeline.
- Orchestrated and optimized data pipelines using scikit-learn, incorporating GridSearchCV for hyperparameter tuning, achieving price prediction accuracy of 91%.

### Cube Logistics

Data Analyst Intern

October 2021 - December 2021

Bengaluru, KA, India

- Enhanced SQL query performance by optimizing SARGability, implementing composite indexing, and restructuring queries, resulting in a 50% reduction in execution times and improved efficiency in a Amazon Redshift-based data warehouse.
- Conducted comprehensive exploratory data analysis on large-scale logistics datasets, uncovering critical KPIs that were strategically integrated into Tableau dashboards, driving enhanced decision-making.

## PROJECTS

### Media Management System|Top 5 at Opportunity Hack|React.js, Postgres, FastAPI, CLIP, 4o-mini, AWS

- Developed and fully deployed a cloud-based system for a NGO within a 24-hour hackathon, implementing AI models (CLIP, 4o-mini) for automated tagging and search, achieving a 92% reduction in manual media organization.
- Engineered a full-stack solution using React.js and FastAPI, integrating AWS services and Dropbox API, enabling seamless media uploads, backups, and scalable search processing.

### Sports Data Processing Pipeline|Python, PySpark, SQL, AWS S3, Apache Spark, Apache Kafka

- Developed a data pipeline leveraging Apache Spark to process cricket match data, handling large-scale data ingestion from Amazon S3, transforming and validating schemas, and optimizing performance for fast querying and analytics.
- Uncovered key insights on player performance and match trends using SQL analytics and visualizations.

### Outlier Detection and Statistical Modeling for Sensitivity-Driven Data Analysis|Python, Statistical Analysis

- Leveraged statistical methods like ANOVA for data preprocessing and classification, while developing and implementing outlier detection models such as MAD and Boxplot, achieving high F-scores (up to 0.99) in detection across diverse datasets.

### Elastic Image Recognition Service on AWS|Python, AWS EC2, ECR, S3, Lambda, SQS, Auto-scaling

- Engineered scalable cloud-based image recognition services using AWS, implementing an advanced auto-scaling algorithm.

## TECHNICAL SKILLS

**Programming Languages** Python, SQL, Shell Scripting.

**Frameworks** PySpark, Flask, Numpy, Pandas, PyTorch, Keras, TensorFlow, Boto3, Scikit-learn, FastAPI.

**Data Technologies** MySQL, PostgreSQL, DynamoDB, Databricks, Snowflake, Apache Spark, Apache Kafka, Apache Spark, Apache Hadoop, Apache Zookeeper, Docker.

**AWS Cloud** AWS EC2, S3, SQS, Lambda, Glue, Athena, QuickSight.

**Skills** System Design, Distributed Systems, Data Structures, Design patterns, ETL, Data Engineering, Data Science, Data Analytics, Deep Learning, Machine Learning, NLP, Generative AI.

**Oracle Certified - Cloud Infrastructure 2024 Generative AI Professional**

**AWS Certified Academy Graduate - AWS Academy Cloud Foundations**